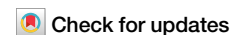


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How tailored climate information affects attitudes towards climate policy and psychological distance of climate change



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How does personally tailored climate impact information affect attitudes towards climate policy paths? We developed an online tool to depict the outcomes of downscaled climate models and a personalised vulnerability estimate. A survey representing Finnish public was conducted to find out the attitudes towards climate policies and psychological distance of climate change among different demographic groups before and after using the tool. We found that tailored information increases knowledge on climate impacts, but the effect on policy attitudes depends on demographic attributes (especially age and income) and related personal vulnerability. After using the tool, the respondents' confidence in their knowledge increased and the potential of climate policies to alleviate changes in the environment was perceived as higher. Most respondents perceived climate impacts as spatially proximal, while temporally and socially there was more variation. For those with low vulnerability levels, climate impacts appeared as more distant after obtaining tailored information.

Climate policies and emission reduction efforts at the national level are essential for the transition to carbon neutrality and for implementing the Paris Agreement. However, the transformation cannot be imposed on a democratic society: it requires broad social support and acceptance, which is built upon public perceptions about the potential risks, needs and fairness of proposed policy paths.

The perception of climate change risks has been considered an important factor explaining climate policy attitudes and the varying levels of support for climate policy^{1,2}. It has been argued that climate impacts remain psychologically distant to many people, which affects the way they perceive the necessity of emission reductions³. Greater perceived distance to climate change consequences has been observed to be linked with less public engagement in climate change issues and less support for climate policies⁴. In addition, the lack of knowledge regarding the variety of consequences that different strategies have leads to tunnel vision (narrow focus e.g., on immediate costs instead of the full spectrum of impacts), which distorts the process of forming an opinion about the fairness of those strategies.

Psychological distance (PD) refers to how distant or proximal a phenomenon is perceived by an individual⁵. PD is constructed by four dimensions: physical, temporal, social and hypothetical. A psychologically distant event affects places far away, occurs far in the future, affects people dissimilar to the perceiver and is uncertain or abstract. Despite the growing occurrence of changes in the living conditions around the world that can be attributed to climate change, several studies have observed that climate

change is still viewed as psychologically distant by people living in countries with comparatively small exposure to climate hazards^{3,6–10}. However, two systematic reviews of studies dealing with PD and climate change find that the evidence is inconsistent^{11,12}.

It has been hypothesised that there is a connection between PD to climate change and climate policy support. This has led to the development of hypotheses about how the reduction of PD by making climate change information better available and more local, temporally close and relevant for individuals might affect the public acceptance of climate policies^{13,14}. Different communication techniques have been proposed to reduce the PD. Loy and Spence¹⁵ suggest *proximising* climate change with news focusing on local impacts, and *bridging* the distance by raising the salience of people's global identity as part of humanity. Van Lange and Huckelba¹⁶ emphasise the need to communicate climate change in a way that transforms it into a concept with concrete consequences and solutions at the local level. Chu and Yang¹⁷ suggest communicating distant and abstract risks by highlighting their disastrous impacts, and communicating concrete risks accompanied with feasible actions. Chu and Yang¹⁸ highlight the need to communicate climate change as something that influences spatially close and familiar things. Breves and Schramm¹⁹ suggest that immersive media is suited to communicate distant risks and that technological immersiveness would create enduring risk perception and enhanced engagement in the topic by simulating direct experiences. van der Linden et al.²⁰ suggest several communication techniques to support societal engagement and climate change policymaking: e.g. emphasising climate change as a present, local, and

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personal risk, and framing policy solutions in terms of what can be gained from immediate action.

Based on their review, van Valkengoed et al.¹² argue that reducing the PD to climate change would not motivate climate action. A study by Miltenberger et al.²¹ emphasises that personalised risk messaging can sometimes indeed reduce concerns over climate change impacts, such as sea-level rise. Schuldt et al.²², who conducted an experiment with U.S. participants, also observed that reduced psychological distance did not translate into increased policy support. Keller et al.¹¹ found inconsistent evidence for the connection between PD and climate attitudes, and suggest researchers of PD to narrow the focus from 'climate change' to specific contexts within it, such as perception of impacts, policy support or sustainable behaviour, in which PD plays a role.

Finland's Climate Act (423/2022) including the carbon neutrality target for 2035 in its Section 2 represents progressive climate policy and calls for widespread public support. Finland and the Nordics have been considered as climate policy pioneers²³, but with ambivalence in the success of policy implementation and output. According to Climate Barometer 2023, a survey which investigated the attitudes of Finnish people towards climate change, a clear majority of Finns (72%) consider that mitigating climate change is an urgent matter. 50% of the respondents contemplate that the Finnish government needs to take more proactive action on climate change mitigation²⁴. Based on a demographically representative survey, Sivonen²⁵ examined the level of public support for global- and national-level climate policy instruments in Finland and the impact of climate risk perception and urban/rural domicile on it. Sivonen observed that support for policy instruments at the global level was typically higher than at the national level, and that higher climate risk perception predicted strong support for certain specific policy instruments (such as national carbon tax). Small differences between urban and rural citizens were found in this study.

There is a gap in prior studies regarding the relationship between climate communication techniques, PD and climate policy support in Finland. Thus, in this study, we investigate the connection of climate impact information to the level of PD among Finnish public and perceptions of climate policy paths. We propose an interactive tool for communicating future consequences of different climate policy paths including personally tailored climate scenario information in the form of different metrics (climatic, health-related, techno-economic) and a personalised vulnerability estimate. We seek to find out whether this information affects the PD to climate change and people's attitudes about alternative climate policy paths. Our expectation is that for climate action it is important to communicate about climate change both in a way that is relevant to individuals (bringing the phenomenon closer) and on a global level (highlighting serious global

consequences). For future work, the study indicates how for some demographic groups, attitudes may be particularly influenced by personal information, while for others, global information may be essential.

This study addresses the identified gap in research regarding PD to climate change, a communication technique for conveying personally tailored climate scenario information, and their connection to climate policy support in Finland by investigating the following research questions:

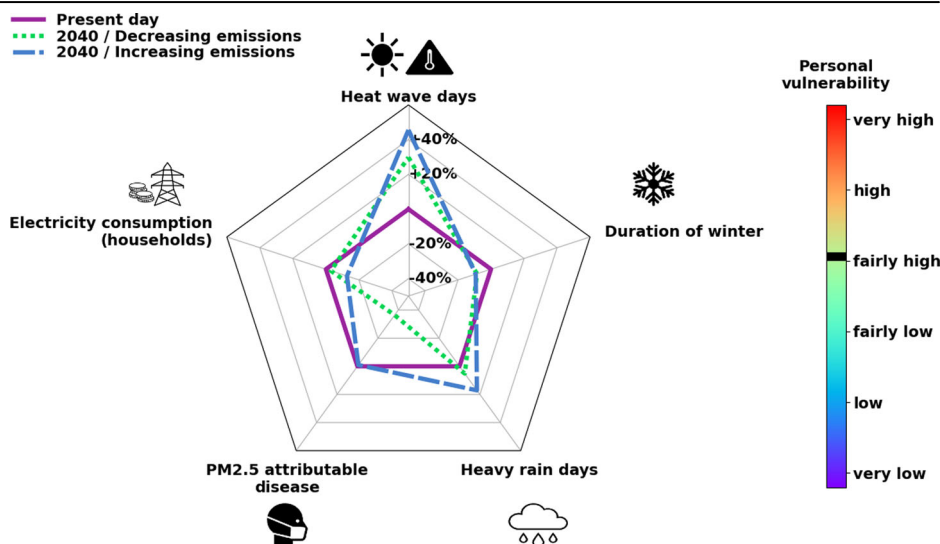
- RQ1 How do Finnish people regard their knowledge level about local climate impacts in different emission scenarios? What is the actual level of knowledge and is it linked to attitudes?
- RQ2 What is the level of psychological distance to climate change in Finland and is there differences between demographic groups?
- RQ3 Does personally tailored climate scenario information affect (a) knowledge, (b) the PD to climate change, and (c) attitudes towards climate policies?

The structure of the rest of the paper is as follows: "Results" describes the results of the study, "Discussion" highlights the conclusions that can be drawn based on the results, and "Methods" introduces the climate scenario data and models that were used, the interactive tool that was developed to convey the scenario data, and the survey that was conducted among the Finnish public.

Results

To address the research questions, this study was built around climate scenarios of high spatial resolution produced with downscaled climate models, an interactive visualisation tool for conveying tailored climate information, and a survey for investigating how the tailored information affects the PD to climate change, knowledge level and attitudes regarding climate policy paths. The results are based on survey responses before and after using the interactive tool (see "Interactive tool" and Fig. 1 for details) to obtain personally tailored information about local climate impacts in two alternative scenarios, one with decreasing emissions (SSP1-2.6) and one with increasing emissions (SSP2-4.5). The number of responses to the survey among Finnish population was 1017. Birth years of the respondents varied between 1942 and 2007 ($Mo = 1971$). The answer options for the domicile of the respondent were given at city or municipality level, and the responses included 142 distinct locations within Finland, 37.6% of which were classified as urban. The level of yearly gross income had eight categories from < 10 000 EUR to > 100 000 EUR, with possibility to decline to respond. All income categories were represented in the sample ($Mo = 30\text{--}40,000$ EUR). According to the survey data, 40.7% of the respondents have a university education, 32.6% of the respondents have

Fig. 1 | An example output of the interactive tool. An example output of the interactive tool showing the changes in five metrics projected to occur at the given location by the year 2040 as compared to the level of present-day in two scenarios, and the person's vulnerability to climatic risks (Source: Clima-teguide.fi web portal and the Finnish Meteorological Institute).



some medical condition relevant for climate risk vulnerability, and 38.9% of the respondents live alone. 49.7% of the respondents were men, 49.8% women, and 0.5% identified themselves as “other”.

Knowledge and tailored climate information

The actual and perceived knowledge level about local climate impacts in different emission scenarios, and whether it could be linked to the attitudes towards climate science and policy, was tested with five claims in connection to knowledge test as listed in Table 4. The knowledge test consisted of two questions: (1) What kind of changes do you expect in your living environment by year 2040, if climate emissions continue increasing?, and (2) What kind of changes do you expect in your living environment by year 2040, if climate emissions start decreasing? For both questions, the respondent was asked to indicate the direction and magnitude of changes they expect regarding the five metrics (described in “Metrics”): will decrease significantly, will decrease slightly, no change, will increase slightly, will increase significantly.

The respondents perceived their knowledge level regarding local climate impacts to be good with 72.5% partly or fully agreeing with the claim I know how climate change will affect me and my living environment by year 2040, if emissions continue increasing before using the interactive tool, and 80.6% after. Responses “Fully agree” increased by 50.5% (+10.5 percentage points) after obtaining the tailored information.

The responses of the knowledge test were compared to actual climate model outputs for the domicile of each respondent and scored. The total average score from the knowledge test was 6.4/20. No significant differences between demographic groups could be found. When combining the knowledge test scores with the respondents’ perceived knowledge level (responses to the claim “I know how climate change will affect me and my living environment by year 2040, if emissions continue increasing”) and their responses to the claim “There is scientific background for the need to mitigate climate change”, we found that the lowest average score (3.5/20) was in the subgroup that strongly agreed with the first claim and strongly

disagreed with the latter claim. High confidence in own knowledge combined with a strong negative attitude towards climate science and policy could thus predict a poor result in knowledge test regarding local climate impacts. Correspondingly, the highest average score (8.0/20) was in the group of respondents who strongly disagreed with the first claim and strongly agreed with the latter claim. Thus, low confidence in own knowledge combined with a strongly positive attitude towards climate science and policy co-occurred with higher than average performance in the knowledge test. Notably, there is a coexistence of these predictions, while the knowledge about climate change impacts and attitudes towards climate policies does not negate the value of self-appraisal of knowledge.

A majority, 81% of the respondents agreed with the claim “The tool was clear and illustrative”. Some ($n = 131$) had left feedback regarding technical challenges with interpreting the graph, such as the tool did not open at all, or the respondents’ firewall preventing the graph from appearing.

Changes in the number of heat wave days by year 2040 was the most relevant piece of information for 34.0% of the respondents. Changes in diseases attributable to air quality were the metric with the most drastic improvement in the scenario with decreasing emissions, and that was regarded as the most relevant piece of information by 25.8% of the respondents.

Changes in electricity consumption was the most relevant piece of information for 19.5% of the respondents, the duration of winter for 14.2%, and changes in the number of heavy rain days for 6.6%.

Attitudes

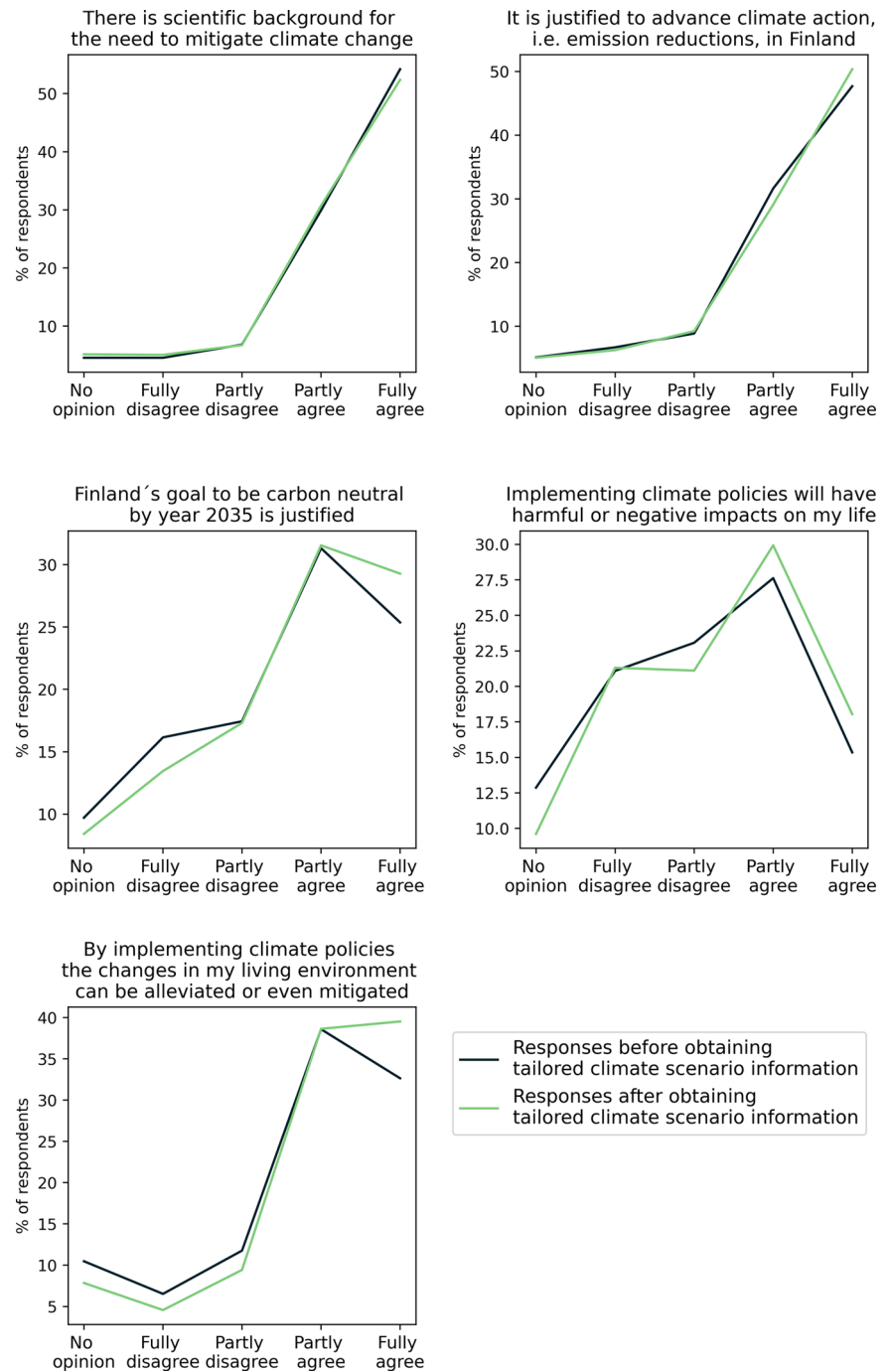
Findings based on responses to the claims testing attitude are summarised in Table 1. Some attributes were associated with responses differing from sample average. The distribution of the responses of the total survey sample to the five claims measuring attitude are shown in Fig. 2.

The popularity of the view “Fully agree” to the claim “There is scientific background for the need to mitigate climate change” correlated with the education level of the respondents.

Table 1 | Findings based on responses to the claims testing attitude

Claim	Attribute associated with responses different from sample average	Finding	Impact of the tool
There is scientific background for the need to mitigate climate change	education	Lowest education: 41.7% fully agree, 12.2% no opinion Highest education: 66.7% fully agree, 0.0% no opinion	none
It is justified to advance climate action, i.e. emission reductions, in Finland	none	-	none
Finland’s goal to be carbon neutral by year 2035 is justified	age	Age > 65: majority fully agree Age < 65: majority partly agree	all: “fully agree” ↑ Age < 20: “fully agree” ↓, partly disagree ↑
Implementing climate policies will have harmful or negative impacts on my life	age	Age > 65: majority fully disagree	Age > 65: no change
	income	Age < 30: majority partly agree	Age < 30: “fully agree” ↑
	domicile type		All: “partly/fully agree” ↑ by 2%-points Income > 70,000 €: “partly/fully agree” ↑ by 11%-points Urban domicile: “fully agree” ↑ Rural domicile: “fully agree” ↓, “partly disagree” ↑
By implementing climate policies the changes in my living environment can be alleviated or even mitigated	age	-	All: “fully agree” ↑ by 6.9%-points
	domicile type		Age < 30: “fully agree” ↑ by 8.5%-points Urban domicile: “fully agree” ↑ by 7.4%-points Rural domicile: no change

Fig. 2 | The effect of the interactive tool on responses in the survey. Distribution of the responses of the total survey sample to five claims measuring attitude towards climate policy. Responses before and after using the interactive tool to obtain personally tailored information about local climate scenarios are shown (Source: Authors).



The claim "It is justified to advance climate action, i.e. emission reductions, in Finland" received similar responses before and after the respondents had used the interactive tool, and no differences between demographic groups or urban vs. rural domicile could be identified.

The claim "Finland's goal to be carbon neutral by the year 2035 is justified" received slightly different responses depending on the demographic group. The fraction of respondents fully agreeing with the claim was the highest among the elderly. In the full sample, the view "Fully agree" increased by 3.9 percentage points with obtaining the tailored scenario information, while among the youngest respondents, the view "Partly disagree" became more popular by 11.4 percentage points. For men and women, the view "Partly agree" was the most popular before using the interactive tool. With using the tool, the views shifted mostly from class "Fully disagree" to "Fully agree", and a larger relative shift towards a more

positive attitude regarding Finland's national carbon neutrality goal occurred among men than among women.

The view "Fully disagree" with the claim "Implementing climate policies will have harmful or negative impacts on my life" was the most popular among the subgroup age > 65 (27%), and unchanged after obtaining tailored climate information. Among the young, the view "Partly agree" was the most popular (31%), and both "Partly agree" and "Fully agree" were reinforced with the information. Agreeing views were reinforced (by up to 11 percentage points) also among the respondents with high income (>70,000 EUR). Thus, age and income level brought differences to how the climate scenarios were regarded. This was expected as those attributes also affect the vulnerability of a person. It could be possible that the young people with low vulnerability conclude e.g. the decreasing number of heat waves in the 'decreasing emissions' scenario as a harmful impact of

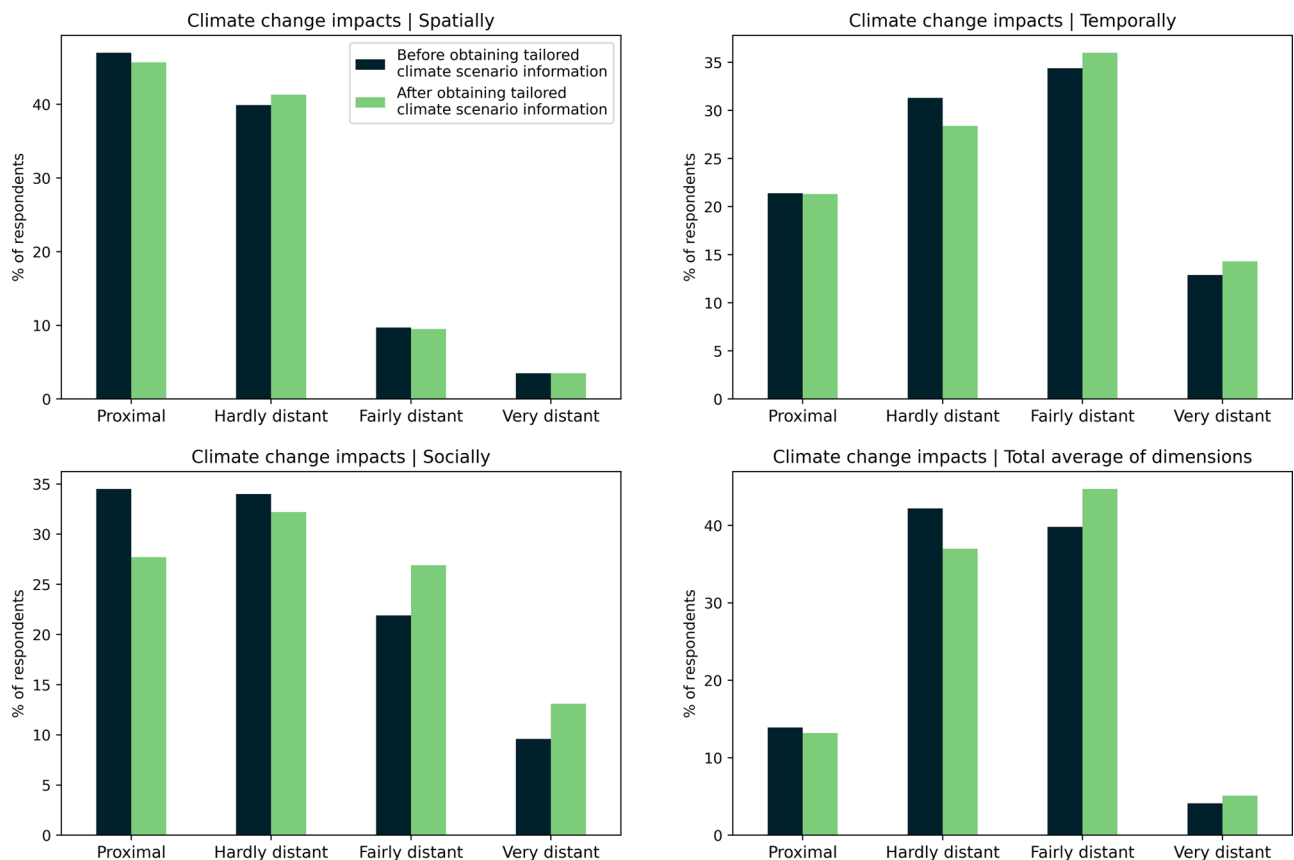


Fig. 3 | The effect of the interactive tool on psychological distance to climate change impacts. The dimensions of psychological distance to climate change impacts as indicated by survey responses. Results for the full sample representing

Finnish adults before and after using the interactive tool to obtain personally tailored information about local climate scenarios are shown (Source: Authors).

climate policies. High income is another factor protecting against climate risks, so a similar mechanism could explain why seeing one's own vulnerability level makes wealthy people agree with the claim. The view "Partly agree" was the most popular for both subgroups urban domicile and rural domicile, but obtaining the tailored information made more respondents with urban domicile fully agree with the claim, while a shift to response class "Partly disagree" was observed among those with rural domicile. Thus, tailored climate information made wealthy and urban respondents regard the impacts of climate policies more negatively, whereas an opposite shift occurred among the rural population.

The majority of respondents in all age groups agreed partly or fully with the claim "By implementing climate policies the changes in my living environment can be alleviated or even mitigated". The view "Fully agree" became more popular after using the tool, and the change away from disagreeing and from having no opinion was accentuated among the young respondents and those with urban domicile. However, this result would partly contrast the interpretation above.

Psychological distance

The different dimensions of psychological distance to climate change were tested with three claims as listed in Table 4. Responses to the claims were converted into indicators of distance at a 4-step scale from proximal to very distant. The distribution of the results based on the total survey sample are shown in Fig. 3. The results for different subgroups are given in the Supplementary Data 1.

Spatially, climate change impacts are deduced to be perceived as proximal or hardly distant by the majority of the respondents. In subgroup age > 65, the majority (more than 50%) of respondent perceptions fell into class "proximal", whereas in other age groups the views were more distributed between classes "proximal", "hardly distant" and "fairly distant".

Only marginal changes in the distribution (mainly from class "proximal" to class "hardly distant") occurred after the respondents had used the interactive tool.

Temporally, climate change impacts appear as fairly distant to 34.4% of the respondents. In subgroup age < 30, a larger fraction than in other subgroups, 43.8%, regards climate change impacts as fairly distant, when it comes to time. In subgroup, age > 65, a larger fraction than in other subgroups, 29.3%, regards climate change impacts as temporally proximal. A slightly larger fraction of all respondents falls into classes "fairly distant" and "very distant" (+1.5 percentage points in each class) after the respondents have obtained tailored climate scenario information regarding year 2040 as compared to today.

A statistically significant shift (paired samples *t*-test, $p < 0.001$) occurs in the perceived social dimension of distance with using the interactive tool. Before obtaining tailored climate scenario information, 34.5% of respondents regarded climate change impacts as socially proximal. With using the tool, a shift of results occurs away from classes "proximal" and "hardly distant" to classes "fairly distant" and "very distant". The social dimension of PD (PD_{social}) increases in all age groups after obtaining the tailored information: by 10% as total average, and at most (by 15%) in the subgroup with birth year between 1990 and 1999.

Regarding subgroups formed based on other attributes than age, it was observed that domicile type, income level and pre-existing medical conditions are relevant for how distant climate change impacts appear socially. Respondents with urban domicile regarded climate change impacts as socially proximal before and after tailored information, whereas those with rural domicile regarded climate change impacts mostly as socially hardly distant in the beginning, and changed their view towards increased distance ($PD_{\text{social}} + 15\%$) after using the tool. Responses in class "Fully agree" to the claim "Climate change will affect the living conditions in Finland, but the

Table 2 | Classification of open responses to question: How did the tailored information about your vulnerability level and the climate impacts on your own living environment affect or change your thinking and perceptions?

Response: change induced by the tailored climate scenario information	Category	Fraction of responses (%)
• More information or reinforcement for my prior knowledge • More accuracy or concrete information about local impacts • I learned that not all impacts are negative • I learned that personal attributes affect my vulnerability • I learned that significant changes may occur within a short time span (by 2040) with increasing emissions	More information/reinforcement for prior knowledge	43.7
• I was surprised that air pollution affects health and that this can be affected with emission reductions • I learned that impacts (on me) are smaller than I thought • I learned that impacts (on me) are larger than I thought • I learned that also I am impacted and/or was surprised that my vulnerability is so high • I was surprised that my vulnerability is so low • I did not expect that also decreasing emissions cause changes in my environment	Surprised by results/Unexpected results	42.4
• I got more motivation to change my behaviour • More positive attitude towards emission reductions • I realised that with emission reductions we could concretely affect our local climate	Motivation, empowerment	7.0
• "It's the end of world" or similar sarcasm • It made me think that it's too late • Increased worry	Emotion	6.1
• It made me think who has ordered this survey and why • There is no point in implementing climate policies - high costs, no benefits	Increased scepticism	0.9

effects will not concern me" increased in all except the highest income group (>70,000 EUR), where the opposite view ("Fully disagree") increased with obtaining the tailored information. Also, climate change impacts were perceived as socially more distant among respondents with basic diseases than among those without. In the latter group, however, the PD_{social} increased by 12% with obtaining the tailored climate scenario information, while in the subgroup with medical conditions the view remained effectively unchanged.

Overall psychological distance to climate change impacts calculated as average of its dimensions increased among the respondents with obtaining the tailored climate scenario information. Results in classes "fairly distant" and "very distant" increased, as can be observed in Fig. 3 which represents the full sample of Finnish adults, and the shift in the distribution was statistically significant (paired samples *t*-test, $p < 0.001$). Total PD to climate change increased with using the tool in all age groups except subgroup *birth year 1950–1959*, in all income groups, in all education level subgroups except the lowest, and in subgroup no basic diseases. For those with urban domicile, total PD increased by 3%, on average, with obtaining the tailored information, while for those with rural domicile the increase in total PD was 11%.

Open responses

The survey included questions with a possibility to give an open response. *Did the tailored information about your vulnerability level and the climate impacts on your own living environment affect your thinking and perceptions? (If so, how?), and What kind of information would you need in order to be able to form your opinion about the legitimacy of alternative climate policy paths?* Based on a data-driven thematic analysis, the responses were classified as shown in Tables 2 and 3. For the first question, there were 328 responses, and for the second, a total of 625 responses. Each individual open response could be classified into more than one category.

The most prevalent responses to the question about changes induced by the tailored information (Table 2) were about receiving more information, getting reinforcement for prior knowledge or obtaining an unexpected result. The respondents were surprised to learn that the impacts on them are so small or so large, and, correspondingly, that their vulnerability is so low or

so high. For 7% of the respondents, the information affected their motivation and attitude towards climate action positively.

The question about information needs (Table 3) obtained a variety of responses. The most common category was "Global perspective" with 26% of responses. The respondents hoped for an opportunity to compare the impacts on Finland and/or Finland's contribution to emission reductions with other countries. The category "An objective truth" received 14.4% of the responses, which indicates that many respondents regarded the provided scenario information as somewhat political or not objective. Concrete specifics and details about climate policies were also hoped for, as well as more information about the impacts on nature in different scenarios.

The responses to the question about information needs align with what has been found previously in related studies. According to a survey ($n = 3806$) on Finnish public's attitudes towards climate change by Tiihonen et al.²⁶, most of the respondents indicated that there is a need for more information on the economic impacts of climate change on Finland (30.2% of the respondents), secondly on a global perspective to climate change mitigation (26.5%) and thirdly on Finland's contribution to global emissions (22.4%). According to the most recent Finnish Science Barometer, a survey about the public opinion and attitudes towards science and technology in Finland, by Varpula²⁷ (N:1085), the majority of Finns trust in science and think that political decision-making should be science-based. However, as per concluded by Varpula²⁷, some tenth of the Finns have indicated only a little trust in science and research, and the influence of politics on scientists' opinions has been indicated as a cause for concern among almost half of the Finns.

Discussion

The research presented in this study was built around climate scenarios with high spatial resolution produced with downscaled climate models, an interactive visualisation tool to convey their results to the public, and a survey for investigating how the tailored information affects the knowledge and attitudes of the Finnish public (a representative sample, 1017 respondents) regarding climate policy paths. The aim of the study was to answer the three research questions detailed in the Introduction.

Table 3 | Classification of open responses to question: What kind of information would you need in order to be able to form your opinion about the legitimacy of alternative climate policy paths?

Response: what kind of information would be needed	Category	Fraction of responses (%)
• Geographic comparison globally/internationally (impacts and/or climate action)	Global perspective, Finland vs. other countries	26.0
• Finland's contribution vs. other countries		
• Discussion on what the big polluters should do		
• Possibility to compare graphs (e.g. two locations) simultaneously		
• Objective/unanimous/unpolitical information	An objective truth	14.4
• The truth (instead of scenarios)		
• Information on costs (for me) and the economic impacts	Information about socio-economic impacts	13.8
• Information on climate refugees and/or conflicts		
• Information on the employment impacts		
• What are the concrete actions behind emission reductions	Details and specifics about actions & impacts	12.2
• The contribution of private cars to climate change		
• How much more electricity would the implementation of emission reductions need		
• How individual climate policies and actions affect the climatic variables		
• Impacts on water systems, sea level and groundwater	Information about the impacts on nature	10.5
• Information about the impacts on biodiversity, plants and wildlife		
• Information about the impacts on agriculture		
• "My opinion does not matter" or "I am not interested"	Nothing (expressed scepticism)	10.5
• Natural variability is behind the changes. Changes would occur even if all emissions stopped.		
• "Less of this"		
• "Just admit that it's all a hoax"		
• An intergenerational view ("the future of the children of today")	Information about fairness	3.9
• Examples of unjust consequences of climate change		
• "What do justice and fairness mean?"	Nothing (a formed opinion)	3.3
• Nothing, my opinion has been formed before		
• A longer time perspective	A different time span	2.2
• A proper wake-up call	Emotional rhetoric	1.1
• Something more appealing to emotions		
• How much would specific diseases such as asthma increase with changing climate	Information about health impacts	1.1
• Information about the positive impacts of climate action	Motivation	1.1

Regarding the first research question on the level of knowledge, the respondents perceived their knowledge about local climate impacts as good, but the knowledge test produced relatively poor results. The lowest scores were achieved by those who strongly believed their knowledge level to be high and had a strong negative attitude towards climate science and policy. Confidence in own knowledge and attitude towards climate information can be concluded to somewhat predict the actual level of knowledge on climate changes.

Interacting with the visualisation tool and obtaining tailored information about local climate-related changes in two alternative scenarios (increasing emissions SSP2-4.5 vs. decreasing emissions SSP1-2.6) caused small shifts in the response distributions indicating attitude towards emission reductions. Overall, opinions got firmer: using the scenario tool reduced the number of responses in class "No opinion". The potential of climate policies to alleviate or even mitigate changes in people's living environment was perceived as higher after obtaining tailored information. However, the effect of tailored information about climate scenario impacts on people's responses was weaker in our study as compared to a previous study by Demski et al.²⁸ exploring the effect of tailored information about future energy pathways.

Our second research question was related to the psychological distance of climate change in Finland and the differences between demographic groups. No significant polarisation in attitudes towards climate science and

policy was found between respondents with urban and rural domiciles. Age, however, was associated with significant differences in attitudes and psychological distance to climate change impacts. In general, the elderly can be concluded to regard climate policies more positively and climate impacts as more proximal than the young do, and the propensity to shift one's view towards a more negative attitude and increased distance with obtaining the tailored information is concluded to be higher among the young.

Based on the results, the Finnish public perceives climate change impacts as spatially proximal or hardly distant, while the temporal and social dimensions of distance indicate more variation and are prone to change due to the conveyed scenario information. Both temporally and socially climate change impacts appeared as more distant after the respondents had familiarised themselves with the scenario information tailored for them. PD_{social} increased especially among the young and healthy, which was somewhat unexpected as younger generations are typically considered to be conscious of climate change.

To answer our third research question about the effect of tailored climate information for PD to climate change and attitudes towards climate policies, we conclude our findings generally that vulnerability factors, such as high age, low-income level, medical conditions, and urban domicile, may lead to experiencing tailored scenario information as something that makes the climate impacts feel personally more relevant and close. For those without such attributes that increase the vulnerability level, tailored scenario

information may increase the psychological distance to climate change impacts. As it has been shown also in the work by van Valkengoed et al.¹² and Mildenerger et al. (2019)²¹, personally tailored climate scenario information is no panacea for reducing PD or for achieving widespread climate policy support. Bringing climate scenario information closer to people, making it consider them as individuals and their local surroundings, has its risks. For some, the information may increase the PD and cause an attitudinal shift towards more negative attitudes about climate policies, while for some the opposite may occur. While previous studies in Finland and beyond have concluded that public concerns over climate change are strongly connected to the public attitudes regarding climate policies^{25,26}, our study further highlights the role of climate information. Our results show that providing tailored information about climate scenarios can increase people's knowledge about climate impacts, but the effect on the attitudes towards climate policies depends on individuals' demographic attributes and their personal vulnerability.

This study contributes to the multidisciplinary research aiming to increase understanding on how climate scenarios are perceived by the public and what affects those perceptions. The concept of the study is reproducible and it would be interesting to see a corresponding study executed in another geographic area with possibly more drastic climate impacts to be conveyed to the public. The limitations of the survey relate to limited resources. Most importantly, the survey did not include a control group, e.g. a group of respondents who would have received general information about climate change on a global level. A control group would have enabled us to analyse the impact of tailored climate information in more detail. Furthermore, the addition of a third subgroup which would have received both tailored and global information, would have enabled an even more sophisticated analysis of the impact of information content in climate change communication. In addition, our aim was to select the most relevant metrics for the visualisation tool, but naturally, the selection could have been done differently which might have led to different results in the survey. For example, 11% of the survey participants would have liked to see more information about the impacts of climate change on nature and biodiversity. However, as the impacts of climate change are expected to be relatively small in Finland when compared with more vulnerable regions, it is highly likely that regardless of the selected metrics, we would have found the increase in PD_{social} due to the low vulnerability of some participants.

Regardless of these limitations, the findings of this study are relevant for climate policy planning and communication. Localised information is useful in climate communication as it brings the topics closer to the daily lives of individuals. However, as low vulnerability is correlated with PD, it is important to also communicate the big picture of climate change and remind people that adverse impacts of climate change in some other places may also be reflected negatively around the world, as we live in a global economy. Moreover, highly vulnerable people are present also in countries with lower risks, such as Finland, thus they should be taken into account when planning climate policies. Whether climate policy paths are viewed as fair or not is relevant to the level of public acceptance, but tapping into its determinants will require an even wider selection of perspectives to be covered. Our recommendation is for future studies to enable the public to obtain comparative results (an opportunity to compare the projected impacts and emission reductions for different locations, countries and globally), and to include more variables describing the impacts of policy paths from different perspectives, such as socio-economic consequences.

Methods

Analysis

This study was designed with quantitative and qualitative research techniques including climate scenarios produced with downscaled climate models, the development of an interactive visualisation tool for conveying tailored climate information, and a survey for investigating how the tailored information affects the PD to climate change, knowledge level and attitudes regarding climate policy paths. The survey analysis consisted of statistical indicators including percentual shares and changes in them, averages and

changes in them, linear regression and paired samples *t*-test. For the open survey responses, a data-driven thematic analysis was conducted to systematically review and categorise the responses based on recurring themes that emerge from the data itself.

Climate models

For local climate scenarios, we applied downscaled global climate model data produced as part of the Climate Impact Lab Global Downscaled Projections for Climate Impacts Research (CIL GDPCIR) project²⁹. In the CIL GDPCIR, CMIP6 model outputs have gone through trend-preserving bias correction and downscaling to achieve a spatial resolution of $0.25 \times 0.25^\circ$. In this study, we obtained gridded data for precipitation and daily minimum and maximum near-surface air temperature from two scenarios: SSP1-2.6 ("Global carbon dioxide emissions experience significant reductions, although at a slow pace. The goal of achieving zero emissions is attained after 2050. The temperature rise is expected to stabilise at approximately 1.8°C by the end of the century"), and SSP2-4.5 ("Carbon dioxide emissions remain relatively steady at present levels before gradually decreasing by the middle of the century. In this scenario, temperatures are projected to increase by 2.7°C by the end of the century.") described by Meinshausen et al.³⁰. The data was accessed via the Microsoft Planetary Computer.

We used an ensemble of source models (global climate models, GCMs): FGOALS-g3³¹, INM-CM5-0³², ACCESS-CM2³³, ACCESS-ESM1-5³⁴, BCC-CSM2-MR³⁵, CMCC-CM2-SR5³⁶, CMCC-ESM2³⁷, EC-Earth3³⁸, GFDL-ESM4³⁹, HadGEM3-GC31-LL⁴⁰, MIROC-ES2L⁴¹, MIROC6⁴², MPI-ESM1-2-HR⁴³, NESM3⁴⁴, NorESM2-MM⁴⁵, UKESM1-0-LL⁴⁶, and CanESM5⁴⁷. Ensemble mean over the listed model outputs was calculated for each variable.

For projected air quality (ambient concentration of particulate matter with an aerodynamic diameter smaller than $2.5\ \mu\text{m}$, $PM_{2.5}$) in the two emission scenarios, we applied the ECHAM-HAMMOZ model^{48,49}. The resolution of the model output is $1.9 \times 1.9^\circ$.

Metrics

The interactive information tool (see "Interactive tool") was designed to depict changes occurring between the present day (year 2020, given as average of 2018–2022) and the scenario year 2040 (average of model outputs for 2035–2045) in the following metrics derived from the climate and air quality model outputs:

Heat wave days. Heat is an interesting variable for the public appraising the local climate, and a health risk for vulnerable demographic groups⁵⁰. The number of heat wave days refers to days that are part of a streak of six days when air temperature exceeds a certain threshold. In Finland, health hazards and risk of mortality increase steeply when a threshold of 22°C is exceeded on consecutive days⁵¹.

Average change by year 2040 in the number of heat wave days in Finland is +18% in the SSP1-2.6 scenario, and +24% in the SSP2-4.5 scenario, based on the ensemble mean of downscaled model outputs.

Heavy rain. The yearly number of days with heavy rain characterises the local climate. It is useful information for e.g. farmers and urban dwellers, and to guide the need for preparation to floods. In this study, heavy rain days were defined as days when precipitation exceeds the threshold of 15 mm.

Average change by year 2040 in the number of heavy rain days in Finland is +14% in the SSP1-2.6 scenario, and +16% in the SSP2-4.5 scenario, based on the ensemble mean of downscaled model outputs.

Duration of winter. The projected duration of winter is relevant information for many sectors and people in a Nordic country. In this study, it was defined as the number of days with sub-zero air temperature.

Average change by year 2040 in the duration of winter in Finland is –5% in the SSP1-2.6 scenario, and –7% in the SSP2-4.5 scenario, based on the ensemble mean of downscaled model outputs.

Air quality attributable diseases. Health issues due to air quality include respiratory and cardiovascular diseases. These health risks (attributable disease, AD) can be numerically evaluated with an exposure-response function that calculates AD as a function of the relative risk of disease (RR_E) at the the long-term average concentration E of an ambient pollutant (here $PM_{2.5}$), and the background disease burden (BoD):

$$AD = \frac{RR_E - 1}{RR_E} \cdot BoD \quad (1)$$

where

$$RR_E = \exp(E \cdot \ln(RR_1)) \quad (2)$$

Risk rate RR_1 in Eq. (2) is specified as $1.062/10 \mu\text{g}/\text{m}^3$ for mortality due to ambient $PM_{2.5}$ ^{52,53}. Disease burden can be numerically expressed in the form of disability-associated life years (DALY). We applied a value of 132,000 per million inhabitants for total BoD expressed as DALY specified for Finnish population, while, for reference, an estimate of 4602 per million for $PM_{2.5}$ specific DALY in Finland has been given⁵⁴.

Average change by year 2040 in the disease burden attributable to $PM_{2.5}$ in Finland is -37% in the SSP1-2.6 scenario, and 0% in the SSP2-4.5 scenario, based on the ECHAM-HAMMOZ simulations.

Electricity consumption for heating in households. The need for heating is a dominant factor for electricity consumption in households. The need is projected to change with the changing climate in Finland by the year 2030⁵⁵. Finnish Meteorological Institute provided modelled scenarios for the need of heating based on simulations by Ruosteenoja and Jylhä⁵⁶.

Based on the model output, the average change in electricity consumption linked to heating demand in Finland by the year 2040 is -3% in the SSP1-2.6 scenario, and -10% in the SSP2-4.5 scenario.

Vulnerability

How climatic risks impact people is dependent on the combination of exposure, i.e. the occurrence of incidences, and people's vulnerability. Bringing climate change information closer to individuals means that personal attributes are taken into consideration and in our study this was done by aggregating them as a vulnerability metric. Vulnerability is a function of a variety of factors including the sensitivity and susceptibility to harm, and the lack of capacity to cope with and adapt to extreme weather and other impacts of climate change. Vulnerability is thus constituted by geographic, environmental and socio-economic variables^{57–59}.

Vulnerability factors considered in this study were: the fraction of artificial surfaces within a radius of two kilometres, flood risk (whether the location has been classified as flood risk area or not), the person's age, income level, education level, pre-existing medical conditions and social isolation. The first two factors are based on location, and the rest are personal or population-level attributes. Each factor gets a numerical value of 1–6 based on the input response, and all the vulnerability factors are aggregated (as geometric mean) into "vulnerability level", which is presented at a scale from very low to very high in the interactive tool (see "Interactive tool"). Here, we aggregated the individual factors with equal weights as has been done in several studies⁶⁰, although finding locally relevant weights for the factors is advisable, and advanced approaches for estimating climate vulnerability are suggested to use reverse weighted geometric aggregation⁶¹.

The land-use data is from CORINE database and the flood risk areas are geospatial data, both available under Creative Commons BY 4.0 International license for open datasets at the Finnish Environment Institute website⁶².

Interactive tool

An interactive tool was developed to communicate the climate scenario information and the vulnerability level described in Sections "Metrics" and "Vulnerability". The tool is available in English at <https://www.climateguide.fi/articles/how-could-climate-change-affect-you/>.

The user inputs their location (e.g. residential area, within Finland), birth year, education, and income level, and answers whether they have pre-existing medical conditions (such as asthma, coronary artery disease or chronic obstructive pulmonary disease) and whether they live alone. After clicking "Generate", the tool creates (1) a radar chart figure depicting the changes in five metrics projected to occur at the given location by year 2040 as compared to the level of present-day in two scenarios, and (2) a colour bar indicating the level of the person's vulnerability to climatic risks. An example of the output graph is shown in Fig. 1. All the information depicted in the graph is also given as text and numbers appearing under the graph in the user interface.

Survey

A national survey representative of Finnish population in regards of age, gender and geographic distributions (only participants with ages > 15 years) was conducted by the company Online Research Finland Ltd as commissioned by the authors. Participants' interest in climate change was not surveyed beforehand, thus it was unknown. Responses were collected during a period from June 29th to July 18th 2023. The survey was in Finnish, and has been translated into English afterwards.

The survey consisted of the following sections:

1. Respondent's basic information
2. Claims about climate change and climate policy paths
3. Knowledge test: changes in the living environment in a scenario of increasing/decreasing emissions
4. Familiarising oneself with the interactive tool and the tailored climate impact information
5. Repeated: Claims about climate change and climate policy paths
6. End questions

All participants used the interactive tool after responding to the first survey, and returned to the second survey immediately after using the tool. The claims of survey sections 2 and 5 are listed in Table 4. Responses were collected on a Likert scale with scoring: fully disagree (1), partly disagree (2), partly agree (3), fully agree (4), no opinion (0). The connection of each claim to the studied aspects (knowledge, attitude, PD) is given in the table.

Survey section 3, the knowledge test, consisted of two questions: (1) "What kind of changes do you expect in your living environment by year 2040, if climate emissions continue increasing?", and (2) "What kind of changes do you expect in your living environment by year 2040, if climate emissions start decreasing?". For both questions, the respondent was asked to indicate the direction and magnitude of changes they expect regarding the five metrics (described in Section "Metrics"): will decrease significantly, will decrease slightly, no change, will increase slightly, will increase significantly. The qualitative responses for the magnitude of change represented numerical changes from $<-30\%$ to $>+30\%$, correspondingly. In the analysis phase, these responses were compared to actual model outputs for the location given by the user, and scored. If the direction of change was correct, the respondent was given one point, and if both the direction and magnitude of change were correct, two points.

Survey section 6, the end questions, included the following questions with possibility to respond with an open response: "Did the tailored information about your vulnerability level and the climate impacts on your own living environment affect your thinking and perceptions? (If so, how?)"; "What kind of information would you need in order to be able to form your opinion about the legitimacy of alternative climate policy paths?"; "Which piece of information given by the interactive tool was the most relevant for you?"; "Was the interactive tool illustrative and clear?"

Table 4 | Claims of the survey and their connection to the studied aspects

Claim	The studied aspect
There is scientific background for the need to mitigate climate change	Attitude, knowledge
I know how climate change will affect me and my living environment by year 2040, if emissions continue increasing	Perception of own knowledge
Climate change will change the conditions in my living environment during the next 20 years	Spatial dimension of PD
Climate change will affect the living conditions in Finland, but the impacts will not affect me	Social dimension of PD
Climate change will affect the living conditions in Finland, but the changes will only occur after several decades	Temporal dimension of PD
It is justified to advance climate action, i.e. emission reductions, in Finland	Attitude towards climate policy
Finland's goal to be carbon neutral by year 2035 is justified	Attitude towards climate policy
Implementing climate policies will have harmful or negative impacts on my life	Attitude towards climate policy
By implementing climate policies the changes in my living environment can be alleviated or even mitigated	Attitude towards climate policy, knowledge

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The authors declare that the main data supporting the findings of this study are available within this article and Supplementary Data 1. The interactive tool which was developed to communicate the climate scenario information and the vulnerability level is available in English at <https://www.climateguide.fi/articles/how-could-climate-change-affect-you/>.

Code availability

The authors declare that the main codes that support the findings of this study are available from the corresponding authors upon reasonable request.

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Author contributions

M.H. did Conceptualisation, Methodology, Formal analysis, Investigation, Writing (Original Draft), Writing (Review & Editing), and Visualisation. T.M. did Conceptualisation, Formal analysis, and Writing (Review & Editing) S.L. took part with Investigation, Writing (Review & Editing), and Project administration. A.L. took part to Conceptualisation, and Writing (Review & Editing) and H.K. was responsible of Supervision, Conceptualisation, and Writing (Review & Editing).

Competing interests

The authors declare no competing interests.

Additional information

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